

HARSHITHA PUTTASWAMY

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PROFESSIONAL EXPERIENCE

Research Assistant, Scalable LLM Inference Framework for Fairness Evaluation

February 2025 – Present

Binghamton University, Department of Computer Science

Binghamton, NY, USA

- Architected and implemented a scalable LLM inference framework leveraging large Language Models to automate the evaluation of envy-freeness in multi-agent item allocations, enabling a consistent benchmarking standard for diverse allocation algorithms.
- Achieved a 4x increase in inference throughput by orchestrating multi-threaded LLM execution across distributed Ollama instances on multiple GPUs, significantly accelerating real-time evaluation of critical fairness properties.
- Engineered advanced prompt engineering strategies to elicit agent preferences, integrating statistical tests to analyze independence between algorithms and personality-driven behavior, validated against MTurk-collected human responses for real-world applicability.

Research Assistant, Reinforcement Learning & Algorithmic Optimization

January 2024 – December 2024

Binghamton University, Department of Computer Science

Binghamton, NY, USA

- Designed and implemented a dynamic Multi-Armed Bandit (MAB) framework to optimize iterative voting decisions, resulting in improved collective utility across agents and effectively modeling sequential decision-making for ad platform bidding strategies.
- Engineered scalable Integer Linear Programming (ILP) algorithms to efficiently solve complex multi-agent resource allocation and ranking problems, enabling strategic and optimized budget distribution within constrained computational environments.
- Conducted extensive experimental analysis on preference aggregation and rule convergence, generating actionable insights for maximizing utility and satisfaction, applicable to designing robust auction mechanisms and real-time bidding environments.

Backend Developer, Internship

March 2022 – December 2022

Fyle Technologies Pvt Ltd

Bangalore, India

- Developed and optimized RESTful APIs using Django, Angular, and PostgreSQL (PSQL), and collaborated with Project Managers to implement 8 new features that increased market competitiveness and user engagement.
- Spearheaded the implementation of a comprehensive CI/CD pipeline using Pytest and GitHub Actions, independently developing tests for 3 core modules that achieved over 98% code coverage each, significantly improving code quality and reducing crash frequency.
- Authored and delivered a knowledge transfer session on Pytest, GitHub Actions, and CI principles, empowering the team to adopt best practices for writing robust tests and understanding the importance of continuous integration for code quality.

PROJECT EXPERIENCE

Immigration Law Chatbot (Fine-Tuned LLM with RAG on AWS) | AWS, Mistral-7B, RAG

May 2025 – June 2025

- Developed and fine-tuned Mistral-7B-instruct LLM for legal Q&A on AWS SageMaker, leveraging a 9,000+ pair custom dataset curated from legal documents using advanced prompt engineering to enhance performance and accuracy.
- Architected and deployed a production-ready, scalable LLM inference system on AWS Lambda and API Gateway, integrating a Retrieval-Augmented Generation system (Amazon OpenSearch) to significantly reduce hallucination and ensure real-world answer relevance.
- Engineered robust and efficient data pipelines for both LLM training and RAG knowledge base, ensuring reliable and seamless ingestion and retrieval of large datasets and source documents from AWS S3.

Splitwiser – A Splitwise clone | Full Stack Development, Amazon Bedrock

June 2024 – December 2024

- Developed a full-stack Splitwise-inspired application, incorporating key features such as receipt scanning, comprehensive group and individual expense tracking, and dynamic graphical/manual split functionalities for enhanced user experience.
- Leveraged Amazon Bedrock API to convert receipt images into JSON data, Next.js for responsive UI and API development, Google Firebase for robust backend services, and Firestore for scalable receipt data storage.

Autokube – Kubernetes Cluster Automation Platform | DevOps, CI/CD

August 2021 – March 2022

- Led the end-to-end development of a web application designed to automate Kubernetes cluster creation and modification, streamlining DevOps workflows.
- Developed robust APIs using the MERN stack (MongoDB, Express.js, React, Node.js) to programmatically initiate scripts, orchestrating complex workflows for efficient management of backend Kubernetes clusters.

EDUCATION

Master of Science in Computer Science

January 2023 – December 2024

Binghamton University, SUNY (State University of New York)

Binghamton, NY, USA

Relevant Coursework: Deep Learning, High Performance Computing, Computer Architecture, Design and Analysis of Algorithms

TECHNICAL SKILLS

Programming Languages: C, CPP, Python, Java, JavaScript, TypeScript, MongoDB, PostgreSQL, MySQL, Firebase

Frameworks and Libraries: TensorFlow, Keras, PyTorch, CUDA, OpenMP, MPI (Familiarity via HPC coursework)

Cloud and Tools: AWS (SageMaker, Bedrock, Lambda, OpenSearch, S3, EC2), Git, Linux, Kubernetes, Github Actions, Pytest

Machine Learning and AI: Recommendation Systems, Pattern Recognition, Deep Learning, Reinforcement Learning, NLP, LLM

Core Computer Science: Data Structures, Algorithm Design & Analysis, Distributed Systems, Computer Architecture

CERTIFICATIONS

AWS Certified Machine Learning Engineer - Associate

- Validates expertise in building, deploying, and managing machine learning solutions on AWS, primarily using Amazon SageMaker.

Reinforcement Learning, Deep Learning Part 1 - IIT Madras (NPTEL) - Elite Certificate

- Covers foundational concepts of deep learning, including various neural network architectures (CNNs, RNNs), optimization algorithms, and attention mechanisms.

CNN in TensorFlow, Neural Networks and Deep Learning, Machine Learning - DeepLearning.AI

- Teaches fundamental concepts of neural networks, including building, training, and applying fully connected deep neural networks with efficient (vectorized) implementations, emphasizing the underlying linear algebra, activation functions, and gradient descent optimization.